21 Mars 2025

Option Learning in IA

Final lab work: Forecasting challenge

Daily data were collected from an *in situ* sensor measuring the level of underground water (in meters). The data are provided in the file *data_train.csv* starting from January the 1st 1995 up to December the 31th 2018. A test file will be used to evaluate your models in the end (*test.csv*).
Your goal will be to provide some models to make forecasting at horizons of 3 months, 6 months and 1 year **with a monthly granularity**.
You have 2 to 3 lab sessions to achieve these goals. You will be evaluated during the last session by presenting shortly your work and intermediary results (steps below). Consequently, at the end of the 2de lab session, you should have at least a full processing chain able to provide some forecasting results (bad or good) at least for 3 months ahead.

You will have to follow these steps:

First Session

1. Analysis of the data (30 min)
    a. Analyze the raw data and draw first conclusions about what you will have to do as preprocessing.
    b. Load the .csv data (with Panda for example) to be able to process them.
    c. Visualize the data respecting the daily scale steps without any preprocessing.
    d. Do any additional analysis you want and draw your conclusions about specificities of the time series and necessary preprocessing to fulfill the expected goal
2. Preprocessing (30 min)
    a. Preprocess and format the series regarding your previous conclusions. **At this step you should have a monthly time series with a value for each month.**
    b. Visualize the new data and control the results
3. First model for 3 months prediction ahead (1h)
    a. Split your data into a training and validation set (2 years)
    b. Choose a model
    c. Make the corresponding ground truth if needed
    d. Build the evaluation tools needed to assess your model from both training set and validation set (ie RMSE and visualization). For validation set, the evaluation should respect 3 months horizon
    e. Fit your model on the training set. If necessary, adjust the model complexity or data complexity (or use a remote platform like Collab) to make it tractable.
    f. Adjust the model/training procedure to improve the results on validation set
    g. Comment on the limits of your approach

Take some rest and think before next session…

Second session

4. Finalize on going task (15 min)
5. At this moment, you should be able to use your model on an unseen file starting from the end of validation set to provide all forecasting values with the given horizon (see test.csv) (15min.)
6. You can try to improve your results or try new models or you can try to go up to next challenge, eg. 6/12 months forecasting (30-60min.)
7. Make everything clean for next session.

Third session

8. The test file is provided. Just run your best model on it and output the results, draw your conclusions. I will check your work during this session. In the meantime, you can try to improve your results on the validation set (not the test ;)